

Creating a UD treebank for Taiwanese Hokkien and Cross-lingual Dependency Parsing from Mandarin Chinese

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Abstract

This paper presents the initial development of a Universal Dependencies (UD) treebank for Taiwanese Hokkien and investigates the feasibility of applying cross-lingual dependency parsing using Mandarin Chinese as the source language. Based on a parallel corpus and manually annotated data from the The Ministry of Education dictionary, we construct a UD treebank for Taiwanese Hokkien and evaluate a cross-lingual model trained on Mandarin Chinese against manually annotated gold-standard data.

1 Introduction

In recent years, Universal Dependencies (UD) has become a widely used framework in Natural Language Processing (NLP). Recently, the UD version 2.16 consists of 296 treebanks in 168 languages. Cross-lingual dependency parsing is also commonly used to automatically annotate low-resource languages.

Taiwanese Hokkien is a Sinitic language spoken by millions of people in Taiwan and Southeast Asia. However, it is not yet included in the UD treebanks. Creating a UD-style treebank for Taiwanese Hokkien and using the cross-lingual parsing from Mandarin Chinese can be an important step toward expanding the UD project and supporting more linguistic research on Sinitic languages.

The Ministry of Education (MOE) in Taiwan provides a dictionary that includes grammatical examples and parallel sentences in Taiwanese Hokkien and Mandarin. Based on this resource, this paper presents a manually annotated small treebank of Taiwanese Hokkien in 200 sentences. In this paper, it also applies a Mandarin Chinese parser to the same sentences and evaluate its output against the gold-standard manual annotations. This allows us to explore the potential of using existing Mandarin parsers to parse Hokkien.

The rest of the paper is organized as follows: Section 2 gives an overview of Taiwanese Hokkien Syntax and the differences between Mandarin and Taiwanese Hokkien Syntax, Section 3 presents the dataset and annotation process, Section 4 details the cross-lingual parsing experiments, and Section 5 discusses the results and future directions.

2 Taiwanese Hokkien Syntax

2.1 The Usage of Taiwanese Hokkien

Taiwanese Hokkien is a Sinitic language spoken in Taiwan and Southeast Asia. According to statistics from the Legislative Yuan ([Legislative Yuan, 2023](#)), based on the 2020 Population and Housing Census, among more than 21 million local residents aged 6 and above, only 31.7% use Taiwanese Hokkien, and the number has been decreasing year by year.

To address this endangered situation, research on Taiwanese Hokkien has been carried out in several directions, including phonetics, syntax, and writing systems. However, there is currently no UD treebank documenting the language. It is therefore necessary to document Taiwanese Hokkien in a systematic way.

Regarding the writing system, Taiwanese Hokkien lacks a fully consistent standard, as it has traditionally been used mainly in spoken contexts. Nevertheless, some orthographies are in use nowadays ([Yu and Tsai, 2008](#)). In this paper, the treebank will primarily adopt Chinese characters as provided in the MOE Taiwanese Hokkien Dictionary, while also including the Taiwanese Romanization System in the MISC field as supplementary information.

2.2 Taiwanese Hokkien Syntax and its Differences from Mandarin

The basic word order of Taiwanese Hokkien is SVO, with the verb typically occurring in the medial position (*Subj + Verb + Obj*). According to

(Wei and Li, 2022), the core grammatical principles of Mandarin and Taiwanese Hokkien are not substantially different. However, several differences remain. One of the most obvious differences lies in the writing systems: the two languages only partially overlap in character usage. For example, characters such as 咧 and 愛 function as auxiliary markers in Taiwanese Hokkien, whereas in Mandarin they do not carry the same grammatical meaning. Similarly, pronouns are represented differently: the third-person singular pronoun in Mandarin is 他, while in Taiwanese Hokkien it is 伊.

Another major difference concerns the syntax of *ADP + Pronoun*. In both languages, oblique elements can sometimes be topicalized, resulting in the pattern *Subj + ADP + Pronoun + Verb* (Wei and Li, 2022), where the verb appears toward the end of the sentence. However, in Taiwanese Hokkien, the pronoun in this construction can function as the subject, object, or oblique, depending on semantic interpretation. In Mandarin, this type of structure is much more restricted and often judged ungrammatical. For instance, in Example 2, Taiwanese Hokkien allows the object to be topicalized, leaving the verb (meaning ‘help’) at the end of the sentence, whereas the equivalent construction in Mandarin would be considered ungrammatical.

(1) Taiwanese Hokkien

我 共 你 講。
guá kā lí kóng
I KA you say
‘I talk to you.’

└ Mandarin

我 跟 你 說。
wǒ gēn nǐ shuō
I with you speak
‘I talk to you.’

(2) Taiwanese Hokkien

我 共 你 鬥 相 共。
guá kā lí tau-sann-kāng
I KA you help
‘I help you.’

└ Mandarin

*我 跟 你 幫 忙。
wǒ gēn nǐ bāngmáng
I with you help
Intended: ‘I help you.’

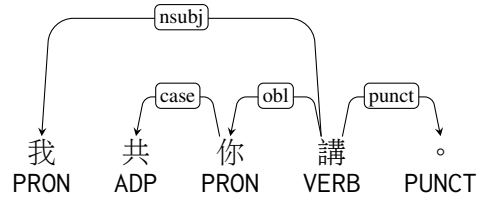


Figure 1: Dependency annotation of the topicalization construction in Example 1.

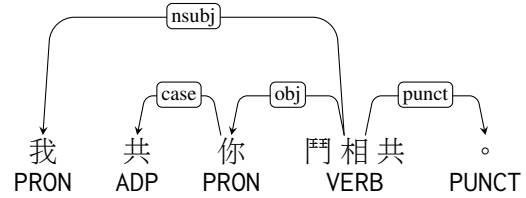


Figure 2: Dependency annotation of the topicalization construction in Example 2.

One more difference in Taiwanese Hokkien syntax is the sequence *Verb + ADP + Pronoun + Complement*. Unlike Mandarin, which often compresses the verb and complement into a compound verb, Taiwanese Hokkien preserves this sequence as a distinct disposal-like construction (Liao, 2007; Wei and Li, 2022; Li, 2005).

(3) Taiwanese Hokkien

肉 愛 煮 予 伊 熟。
bah ai tsu hoo i siok
meat want cook KA him ripe
‘The meat needs to be cooked for him until it is done.’

└ Mandarin

*肉 要 煮 給 他 熟。
rou yao zhu gei ta shu
meat want cook GEI him ripe
Intended: ‘The meat needs to be cooked for him until it is done.’

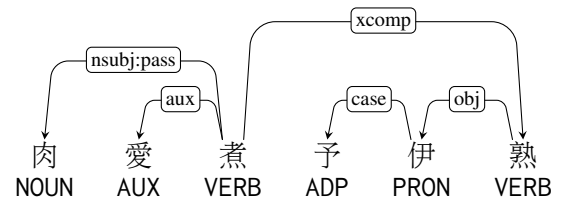


Figure 3: Dependency annotation of the disposal construction in Example 3.

Therefore, the hypothesis of using Mandarin as the source language for cross-lingual dependency

parsing into Taiwanese Hokkien appears highly plausible, since the main syntactic structures are similar. However, several challenges remain, including characters shared with Mandarin but carrying different meanings in Taiwanese Hokkien, the frequent use of complement clauses, and the topicalization of pronouns, which in Taiwanese Hokkien can function as obliques, subjects, or objects depending on the semantic context.

3 Manual Annotation

In this project, 200 sentences of Taiwanese Hokkien were annotated in the Universal Dependencies (UD v2) framework. The sentences were mainly drawn from the MOE’s Taiwanese Hokkien Dictionary and supplemented with colloquial examples, in order to capture both written and spoken usage. Each token was segmented in Chinese characters, and the corresponding Taiwanese Romanization (Tâi-lô) form was added in the MISC field as additional information.

A major challenge was word segmentation, since Taiwanese Hokkien lacks a standardized orthography and many multi-character expressions can be interpreted either as single lexical units or as compositional phrases. For example, compound verbs such as 轉去 (trng-khì, ‘come back’) or compound nouns like 衫仔褲 (sann-á khòo, ‘clothes and pants: means clothing’) raise the question of whether they should be treated as one word or split into conjunction. In these cases, consistency with existing UD treebanks (especially Mandarin GSD) was taken into account while keeping the specific grammatical properties of Taiwanese Hokkien.

For part-of-speech tagging and dependency relations, the UD v2 guidelines were followed, with reference to the Mandarin UD (GSD) treebank to ensure cross-lingual comparability. Nevertheless, several adaptations were required. For example, aspectual markers such as 咧 (leh) were treated as auxiliaries, while multifunctional items like 無 (bô) were annotated differently depending on whether they functioned as a verb (‘not have’) or an adverb (‘not’), following the categorizations provided in the MOE Taiwanese Hokkien Dictionary. A special focus was given to the disposal construction (V + 予 hoo + pronoun + complement), which is productive in Taiwanese Hokkien but absent in Mandarin, where the verb and complement are often fused into a compound verb. Table 1 shows the distribution of dependency relations: xcomp accounts for

4.39% and ccomp 1.54%, both relatively high.

Another recurring challenge was deciding the dependency relations between clauses. In some cases the relation could be annotated as *advcl*, while in others it was better captured as *parataxis*, depending on the semantic relation between the clauses.

The annotation process highlights three main challenges: the lack of standardized segmentation, mismatches between Mandarin-based UD conventions and Hokkien-specific syntax, and deciding between *acl* and *parataxis* for clausal relations. Nevertheless, the structural similarities between Mandarin and Taiwanese Hokkien suggest that this annotated dataset may provide a useful foundation for future cross-lingual parsing experiments.

4 Cross Lingual Parsing

4.1 Experimental Setup

The cross-lingual dependency experiment in this project was partly inspired by the methodology of (Zhou et al., 2025). Unlike their approach of training a parser for the source language, we relied on the pre-trained parser Stanza, provided by Stanford University.

To assess how the parser performs on Taiwanese Hokkien, we conducted three different experimental settings:

1. **Full pipeline:** letting Stanza handle the entire process, including tokenization, POS tagging, and dependency parsing.
2. **Tokenization only:** providing the parser with gold tokenization, while leaving POS tagging and dependency parsing to Stanza.
3. **Tokenization + POS tagging:** supplying both gold tokenization and gold POS tags, and letting Stanza only perform dependency parsing.

For all three versions, we directly fed Taiwanese Hokkien sentences into the parser. The outputs were then systematically compared against our manual UD annotation of 200 sentences. This comparison allows us to examine whether and to what extent the parser diverges from human annotation, and to evaluate the feasibility of applying Mandarin-trained parsers to Taiwanese Hokkien.

4.2 Results

For the full pipeline experiment, we observed that the parser suffered from tokenization mismatches

Deprel	Count	Percentage
advmod	302	14.57
nsubj	243	11.72
root	201	9.70
obj	131	6.32
aux	123	5.93
xcomp	91	4.39
obl	83	4.00
advcl	81	3.91
case	79	3.81
nmod	77	3.71
mark	55	2.65
det	43	2.07
discourse	43	2.07
ccomp	32	1.54
clf	31	1.50
parataxis	29	1.40
nummod	28	1.35
acl	18	0.87
compound:dir	16	0.77
amod	16	0.77
conj	13	0.63
cop	9	0.43
appos	8	0.39
csubj	7	0.34
obl:agent	5	0.24
cc	5	0.24
nsubj:pass	4	0.19
compound:vv	3	0.14
compound:vc	2	0.10
compound	2	0.10
iobj	1	0.05
acl:relcl	1	0.05
vocative	1	0.05

Table 1: Distribution of dependency relations in Manual Annotation.

compared to the gold standard, which posed challenges for evaluation. As a result, UAS and LAS were calculated only on the sentences where the tokenization aligned with the gold annotation, and these values serve as the baseline for comparison.

When providing tokenization based on the gold annotation, both UAS and LAS showed a substantial improvement in performance. The improvement was further amplified when both tokenization and POS tags were supplied. A comparison of the three setups is summarized in Table 2.

Since tokenization differs in the full pipeline version, Table 3 only reports the evaluation results for

the gold tokenization and gold tokenization with POS tags settings. We observe that the correctness of subjects, objects, and obliques improves noticeably when gold POS tags are provided. In addition, for complement clauses—which frequently occur in Taiwanese Hokkien—the correctness also shows a substantial improvement once the parser has access to POS information.

However, the correctness of xcomp remains relatively low. Moreover, the accuracy of cop and parataxis decreases when POS information is provided to the parser. This fluctuation may be due to the limited size of the dataset (only 200 sentences), which does not provide sufficient data for stable evaluation. Overall, however, the results on this small corpus indicate that when using Mandarin as the source language and supplying gold tokenization and POS tags, the correctness of dependency relations and the UAS both improve, suggesting the potential feasibility of cross-lingual parsing from Mandarin to Taiwanese Hokkien.

5 Conclusion

This study presented a first attempt at constructing a small Universal Dependencies (UD) treebank for Taiwanese Hokkien and testing the feasibility of cross-lingual dependency parsing using Mandarin as the source language. The experiments compared three setups: a full pipeline, gold tokenization, and gold tokenization with POS tags. The results show that providing gold tokenization and POS tags yields substantial improvements in parsing accuracy, especially for core syntactic relations such as subjects, objects, and obliques. However, some relations such as xcomp, cop, and parataxis remain difficult, partly due to the limited dataset size. Overall, the findings provide preliminary evidence that cross-lingual parsing from Mandarin to Taiwanese Hokkien is feasible, provided that basic annotation layers (tokenization, POS tags) are available.

Future work can proceed in several directions. First, expanding the annotated dataset beyond the current 200 sentences is crucial for more robust evaluation and for covering a wider range of syntactic phenomena. Improving tokenization is another key step: this could involve establishing a stable segmentation standard or exploring subword- and character-level models to mitigate inconsistencies. Second, instead of relying solely on an off-the-shelf Mandarin parser, one promising direction is to fine-tune a Mandarin parser on Taiwanese Hokkien

Setup	UAS	LAS
Full pipeline	19.53	14.46
Gold tokenization	40.23	26.58
Gold tokenization + POS	58.32	44.77

Table 2: UAS and LAS performance across different experimental setups.

Deprel	Gold tokenization			Gold tokenization + POS		
	Prec	Rec	F1	Prec	Rec	F1
nsubj	0.28	0.36	0.31	0.50	0.56	0.53
aux	0.52	0.41	0.46	0.87	0.61	0.72
root	0.33	0.33	0.33	0.51	0.50	0.50
ccomp	0.18	0.38	0.24	0.30	0.41	0.34
obj	0.41	0.46	0.43	0.65	0.50	0.56
advmod	0.44	0.14	0.21	0.75	0.62	0.68
parataxis	0.11	0.14	0.12	0.07	0.03	0.05
xcomp	0.08	0.08	0.08	0.21	0.25	0.23
cop	0.69	1.00	0.82	0.26	1.00	0.41
case	0.50	0.30	0.38	0.29	0.78	0.42
nummod	0.42	0.54	0.47	0.39	0.61	0.47
det	0.32	0.21	0.25	0.46	0.30	0.37
obl	0.09	0.01	0.02	0.27	0.19	0.23
mark	0.05	0.09	0.06	0.17	0.09	0.12
advcl	0.06	0.11	0.08	0.10	0.06	0.08
nmod	0.15	0.19	0.17	0.27	0.45	0.34
clf	0.50	0.35	0.42	0.62	0.26	0.36
appos	0.05	0.12	0.07	1.00	0.12	0.22

Table 3: Comparison of dependency relation-wise accuracy (Precision, Recall, F1) between gold tokenization and gold tokenization + POS.

data so that it can adapt to language-specific constructions while leveraging structural similarities between the two languages.

Resources

The annotated UD treebank for Taiwanese Hokkien (200 sentences) and the code for cross-lingual dependency parsing experiments are publicly available on GitHub: <https://github.com/Yi-Shan-lin/th-ud-treebank-dtdp.git>.

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